higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CARPENTRY AND ROOF WORK
NQF LEVEL 2

22 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1: BASIC TOOLS AND TERMINOLOGY

1.1 1.1.1
- Body
- Bubble tubes (2)

1.1.2 A spirit level is used for testing whether a surface is level or plumb. (1)

1.1.3
- Water-pipe level
- Plumb bob (2)

1.2
- A crowbar has more leverage.
- A crowbar can remove nails with less effort. (Any ONE) (1)

1.3 Advantage: Do not need electrical power to work
Disadvantage: Time consuming/Requires more effort (2)

1.4
- Wear ear protection.
- Read all warning labels and operating instructions.
- Do not wear loose-fitting clothes as they may get caught in the machine.
- Wear eye protection. (Any 3 × 1) (3)

1.5
- Valley truss
- Scissors truss
- Girder truss
- Truncated truss
- Jack truss
- Howe truss
- Hip truss (Any 5 × 1) (5)

1.6 Batten size: 38 mm × 38 mm✓ support roof tiles✓
Purlin size: 76 mm × 50 mm✓ support roofing sheets✓ (2 + 2) (4)

[20]

QUESTION 2: ROOF CONSTRUCTION

2.1 Covers and protects the interior of a building and its occupants (1)

2.2
- Live loads
- Dead loads (2)

2.3
- It is self-bracing and requires less diagonal bracing.
- It improves the appearance of the building.
- It allows the roof designers to be innovative. (3)

2.4 Rationally designed roofs✓ are designed by engineers using scientific calculations,✓ while deemed-to-satisfy roof designs✓ are developed by craftsmen based on experience.✓ (2 × 2) (4)

2.5
- Tilting battens
- Safe slinging and lifting of a truss
- Transporting, handling, storing and levelling of trusses
- Different types of hurricane clips
- Correct installation of battens and purlins
- Effective use of roof anchors
- Right wall plates for different roof spans
- Correct spacing of trusses for the type of roof covering (Any 5 × 1) (5)

2.6 A wall plate is a length of timber that lays flat on a supporting wall✓ on which the roof trusses rest.✓ The wall plate is used to make the bearing surface of the wall level✓and helps to spread the weight of the roof evenly.✓ (4)

2.7 When the pitch of the roof is very steep (more than 45°) (1)

2.8
2.8.1 False
2.8.2 False
2.8.3 False
2.8.4 True
2.8.5 True
(5 × 1) (5)

2.9
- Roof anchors must be fixed tightly so that the trusses do not move.
- Roof anchors must not be more than 200 mm away from the truss. (2)

2.10 Step 1: Bend the hoop iron over the top chord.
Step 2: Pull the hoop iron tightly over the top chord.
Step 3: Return the hoop iron down the opposite side of the chord.
Step 4: Nail the hoop iron securely over the top chord of the truss. (4)

2.11 Ring shank nail (1)

2.12
- The joints between the truss members will remain tight.
- Damage or weakening of the trusses will be prevented.
- Trusses will not lose their shape. (3)

2.13
- Trusses can be constructed using only hand tools.
- They can be constructed on site. (2)

2.14
- Washers improve the holding capacity of the bolts.
- Washers prevent the head of the bolt from pulling through the timber. (2)

2.15
- Spreader bar
- Belt sling
- Strong back (Any ONE) (1)

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QUESTIONS 3: BUILD A HOWE ROOF TRUSS

3.1
- Set out truss (draw template).
- Lay the timber truss members on the template and cut them to size.
- Assemble the truss using nails and other connectors. (3)

3.2
3.2.1 Run
3.2.2 Pitch
3.2.3 Apex
3.2.4 Overhang
3.2.5 Clear span
3.2.6 Ridge

(6 × 1) (6)

3.3
- Shoot a horizontal line for the tie beam.
- Mark the centre point of the line A.
- Measure off 300 mm horizontally from the centre point B.
- Scribe an arc of 400 mm from the centre point perpendicular to the tie beam line.
- Scribe an arc of 500 mm from the 300 mm mark on the tie beam line to intersect with the 400 mm arc C.
- Join the intersection point of the two arcs C to the centre point A on the tie beam line.
- The angle BAC should be equal to 90° (square). (8)

3.4
- Struts
- Webs (1)

3.5 Nail/Gang-nail plate (1)

3.6 Sliding bevel (1)

[20]

QUESTION 4: GABLE ROOF CONSTRUCTION

4.1
4.1.1 C
4.1.2 D
4.1.3 C
4.1.4 C
4.1.5 D
4.1.6 B
4.1.7 B
4.1.8 C
4.1.9 B
4.1.10 D

(10 × 1) (10)

4.2
I
C
B
E
G
D
J
F
H
A

(10 × 1) (10)
[20]

TOTAL: 100

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